Warren Li, PhD

Learning Analytics and AI Researcher, Educator, and Systems Builder

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Industry Experience

May 2026 – Present
Research Scientist, Learning Data Insights
Conducted a literature review to develop the technical requirements for a $5.5M grant program funding open-source K-12 math AI evaluation benchmarks.

Led technical diligence across a varied portfolio of 10+ major EdTech organizations funded by a large nonprofit; reviewed and strengthened research portals and documentation to increase the visibility of digital public goods.

May 2024 – May 2026
Associate Research Scientist, Learning Data Insights
Developed a hybrid human-AI rubric-refinement method for automated scoring across 198 student programming assignments using OpenAI reasoning models, reaching 95% agreement with human raters on select constructs.

Identified and procured a proposal submission platform and organized the review process for the EduData Initiative; built and curated the Digital Learning Platforms (DLP) Catalog (>2,600 views, 400 users) to promote sharing of data dictionaries/mock data.

Co-developed a user interface to strip personally identifiable information from classroom transcripts using LLMs and local open-weight models, enabling privacy-preserving education research while maintaining context with greater reliability than rule-based approaches.

Built and deployed a data pipeline and dashboard (R Shiny on AWS) unifying IES and USAspending.gov records into a consolidated view of ∼$2.5B in federal education R&D funding for policymakers and the public.

January 2015 – May 2015
Software Developer Intern, World Wide Technologies (WWT): Asynchrony Labs
Used Agile methods to develop Qwizard, a responsive Angular and Node.js single-page web application for internal company communications; organized software demonstrations for stakeholders.

Research Experience

May 2018 – April 2024
Graduate Student Research Assistant (GSRA), University of Michigan: School of Information

1 **Biases in Predictive Modeling and Understanding Privacy Perceptions of Students**

Conducted large-scale survey with 4,000 students and two experimental groups to understand key privacy concerns and implications on educational research when faced with legal restrictions (e.g. FERPA, GDPR).

Invited for an interview for Grow with Google and University of Michigan's course "Data Analytics in the Public Sector with R".

2 **Explainable AI Learner Models and its Impact on Decision Making when Learning**

Built a Flask app hosted on Google Cloud for students to upload Jupyter Notebooks, make self-evaluations, and compare with perceived AI annotations to log metacognitive behavioral data.

Published with a psychometrician at the Michigan Department of Education on innovative methods using NLP and ML techniques to review technology-enhanced test items.

June 2016 – August 2016 **Nuclear Physics Research Intern**, Wayne State University: Department of Physics and Astronomy

Selected for the NSF Research Experiences for Undergraduates (REU) Program. Wrote and presented report analyzing large datasets from simulated collision events to study variation of collision vertex and experimental inefficiencies on rapidity correlations.

Teaching Experience

January 2021 – April 2021 **Lecturer/Instructor of Record**, University of Michigan: School of Information

Led SI330: Data Manipulation for 61 upper-level students, developed syllabus incorporating gamified elements for numpy/scipy and recorded and edited video lectures on YouTube.

Managed a team of two instructional aides and one graduate student instructor; oversaw the new design, administration, and grading of a scalable oral exam for distance learning.

Sep. 2020 – Dec. 2020 **Graduate Student Instructor (GSI)**, University of Michigan: School of Information

Hosted office hours and supplementary tutorial sessions for 42 students; nominated for GSI of the Year Award, with teaching evaluations above university median for all metrics.

Designed collaborative active-learning code tutorials from scratch using JupyterHub during discussion sections to maintain student interactions during the pandemic online shift.

January 2019 – June 2019 **Course Content Assistant (CCA)**, University of Michigan: School of Information

Created all new lecture scripts, assignments, and curriculum (pandas/matplotlib) for the Visual Exploration of Data (VED) course launch in the Master's of Applied Data Science (MADS) online degree program for over 600 students in the inaugural cohort.

October 2016 – May 2017 **Teaching Fellow**, The LaunchCode Foundation

Created and shared publicly-accessible lesson plans and supplemental projects for LaunchCode, a nonprofit organization providing free and accessible tech training.

Volunteered over 50 hours outside of class responding to discussion boards, conducting study sessions (Python and Google App Engine); helped multiple students land their first positions in tech.

August 2015 – May 2017 **Mathematics Tutor**, University of Missouri - St. Louis: Math and Writing Academic Center

Instructed students on a walk-in basis with algebra, probability and statistics, calculus, differential equations, linear algebra, and discrete math. Also provided one-on-one tutoring through the Office of Multicultural Student Services (MSS) with a 100% pass rate.

Education

2018–2024 **PhD in Information**, *School of Information, University of Michigan, Ann Arbor*

Thesis: Understanding Learners and The Interplay Between Metacognitive Judgements of Learning and AI-Generated Explanations

2017 **BS in Engineering Physics, BA in Mathematics, Minor in Computer Science**, *University of Missouri, St. Louis*
Summa Cum Laude (top 3%)

Peer-Reviewed Publications

1 Zeng, J., **Li, W.**, & Heck, T. (2025). *Visualizing Item Statistics for Better Developing Technology-Enhanced (TE) Items*. Applied Measurement in Education, 38(3-4), 331-347.

2 **Li, W.**, & Brooks, C. (2024). *Influence on Judgements of Learning Given Perceived AI Annotations*. In Proceedings of the Eleventh ACM Conference on Learning @ Scale (pp. 221-231).

3 Singh, A., Brooks, C., Wang, X., **Li, W.**, Kim, J., & Wilson, D. (2024). *Bridging Learnersourcing and AI: Exploring the Dynamics of Student-AI Collaborative Feedback Generation*. In Proceedings of the 14th Learning Analytics and Knowledge Conference (pp. 742-748).

4 Choi, H., Winne, P. H., Brooks, C., **Li, W.**, & Shedden, K. (2023). *Logs or Self-Reports? Misalignment Between Behavioral Trace Data and Surveys When Modeling Learner Achievement Goal Orientation*. In LAK23: 13th International Learning Analytics and Knowledge Conference (pp. 11-21).

5 Motz, B., Quick, J., Brooks, C., Bergner, Y., Gray, G., Lang, C., **Li, W.** & Marmolejo-Ramos, F. (2023). *A LAK of direction: Misalignment between the goals of learning analytics and its research scholarship.* Journal of Learning Analytics, 10(2), 1-13.

6 **Li, W.**, Sun, K., Schaub, F., & Brooks, C. (2022). *Disparities in Students' Propensities to Consent to Learning Analytics.* International Journal of Artificial Intelligence in Education. 32(3), 564-608.

7 Singh, A., Brooks, C., Lin, Y., & **Li, W.** (2021). *What's In It for the Learners? Evidence from a Randomized Field Experiment on Learnersourcing Questions in a MOOC.* In Proceedings of the Seventh ACM Conference on Learning @ Scale (pp. 221-233).

8 Nguyen, Q., Poquet, O., Brooks, C., & **Li, W.** (2020). *Exploring Homophily in Demographics and Academic Performance Using Spatial-Temporal Student Networks.* In Proceedings of The 13th International Conference on Educational Data Mining (EDM 2020) (pp. 194-201).

9 **Li, W.**, Brooks, C., & Schaub, F. (2019). *The Impact of Student Opt-Out on Educational Predictive Models.* In Proceedings of the 9th International Conference on Learning Analytics & Knowledge (pp. 411-420). ACM.

10 Gardner, J., Brooks, C., & **Li, W.** (2018). *Learn from Your (Markov) Neighbour: Co-Enrollment, Assortativity, and Grade Prediction in Undergraduate Courses*. Journal of Learning Analytics, 5(3), 42-59.

Talks and Presentations

1 **Li, W.**, Kurimchak, M., Andres, J.M.A.L., & Whitmer, J. *Evaluating the Landscape of Automated Scoring with GenAI: A Systematic Review*. To be presented at the Impactful and Responsible AI Systems for Education (IRAISE) Workshop - Festival of Learning. Seoul, South Korea. June 28, 2026.

2 **Li, W.**, Rutstein, D. & Whitmer, J. *Hybrid Human-AI Rubric Development for Interpretable Automated Evaluation in Computer Science Assessments*. Presented at the Annual Meeting of the National Council on Measurement in Education (NCME). Los Angeles, CA. April 11, 2026.

3 **Li, W.** & Brooks, C. *Influence on Judgements of Learning Given Perceived AI Annotations*. Presented at the 11th ACM Learning @ Scale Conference. Atlanta, GA. July 18, 2024.

4 **Li, W.** & Brooks, C. *Use of Open Learner Models During Metacognitive Calibration*. Presented at the 14th International Conference on Learning Analytics and Knowledge (LAK). Kyoto, Japan. March 18, 2024.

5 **Li, W.**, Sun, K., Schaub, F., & Brooks, C. *Disparities in Students' Propensity to Consent to Learning Analytics*. Presented at the 23rd International Conference on Artificial Intelligence in Education. Durham, UK. July 30, 2022

6 **Li, W.** *Metacognitive Judgements and its Influence on Decision-Making with Machine-Generated Explanations*. Presented at the 12th International Conference on Learning Analytics & Knowledge (LAK) Doctoral Consortium. Virtual. March 22, 2022.

7 **Li, W.**, Brooks, C., & Schaub, F. *The Impact of Student Opt-Out on Educational Predictive Models*. Presented at the 9th International Conference on Learning Analytics & Knowledge (LAK). Tempe, AZ. March 7, 2019

Service

2021, 2023 Program Committee member for the International Journal of Artificial Intelligence in Education

2018, 2022 Program Committee member for the Learning Analytics and Knowledge Conference (LAK)

2017 First annual University of Missouri - St. Louis Hackathon (UMSLHack) volunteer mentor for the 3rd place team

Grants and Fellowships

2019, 2022 UMSI Travel Grant ($700)

2019, 2022 Rackham Travel Grant ($800)

2023 Rackham Graduate Student Research Grant ($2,000)

2019 National Science Foundation: Graduate Research Fellowship ($138,000 across 3 funding years, selectivity rate $< 15\%$)

Languages

Cantonese Speaking/Listening: Conversational (B2), Reading/Writing: Intermediate (B1)

Mandarin Speaking/Listening: Conversational (B2), Reading/Writing: Intermediate (B1)

French Professional Proficiency (Test de connaissance du français - TCF: B2)

English Native/Fluent Proficiency (C2)